17. PACELC Theorem

Let's learn about the PACELC theorem and its usage.

Background

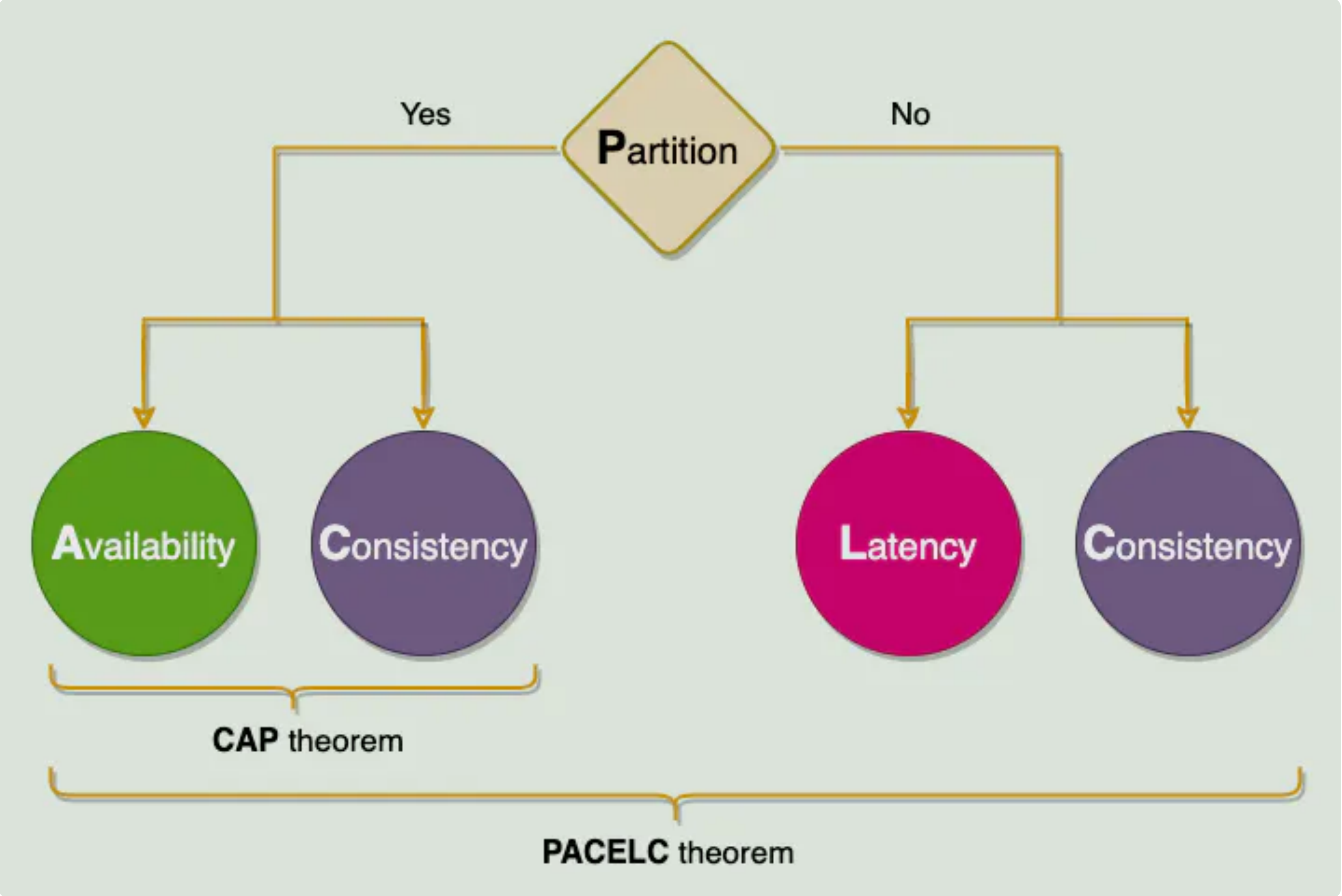
We cannot avoid partition in a distributed system, therefore, according to the CAP theorem, a distributed system should choose between consistency or availability. **ACID** (Atomicity, Consistency, Isolation, Durability) databases, such as RDBMSs like MySQL, Oracle, and Microsoft SQL Server, chose consistency (refuse response if it cannot check with peers), while **BASE** (Basically Available, Soft-state, Eventually consistent) databases, such as NoSQL databases like MongoDB, Cassandra, and Redis, chose availability (respond with local data without ensuring it is the latest with its peers).

One place where the CAP theorem is silent is what happens when there is no network partition? What choices does a distributed system have when there is no partition?

Definition

The PACELC theorem states that in a system that replicates data:

- if there is a partition ('P'), a distributed system can tradeoff between availability and consistency (i.e., 'A' and 'C');
- else ('E'), when the system is running normally in the absence of partitions, the system can tradeoff between latency ('L') and consistency ('C').



PACELC theorem

Solution

The first part of the theorem (**PAC**) is the same as the CAP theorem, and the **ELC** is the extension. The whole thesis is assuming we maintain high availability by replication. So, when there is a failure, CAP theorem prevails. But if not, we still have to consider the tradeoff between consistency and latency of a replicated system.

Examples

- **Dynamo and Cassandra** are **PA/EL** systems: They choose availability over consistency when a partition occurs; otherwise, they choose lower latency.
- **BigTable and HBase** are **PC/EC** systems: They will always choose consistency, giving up availability and lower latency.
- **MongoDB** can be considered **PA/EC** (default configuration): MongoDB works in a primary/secondaries configuration. In the default configuration, all writes and reads are performed on the primary. As all replication is done asynchronously (from primary to secondaries), when there is a network partition in which primary is lost or becomes isolated on the minority side, there is a chance of losing data that is unreplicated to secondaries, hence there is a loss of consistency during partitions. Therefore it can be concluded that **in the case of a network partition, MongoDB chooses availability, but otherwise guarantees consistency**. Alternately, when MongoDB is configured to write on majority replicas and read from the primary, it could be categorized as PC/EC.

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